A SEQUENCE THAT IS REGULAR AFTER LOCALISING AT A PRIME STAYS REGULAR AFTER INVERTING ONE ELEMENT OUTSIDE IT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The kernels whose vanishing is regularity. Defined in the counted proof at lines 16923-16927. Encode the regularity of the sequence as the vanishing of finitely many kernels of multiplication.
- A finite module vanishing after localisation vanishes after inverting one element. Used at the last two lines of the counted proof (lines 16928-16930). Turns the vanishing of the localised kernels at the given prime into vanishing after inverting some g outside that prime.

Lemma 0.1. *Let R be a Noetherian ring. Let M be a finite R-module. Let $\mathfrak{p}$ be a prime. Let $x_1, \ldots, x_r$ be a sequence in R whose image in $R_\mathfrak{p}$ forms an $M_\mathfrak{p}$-regular sequence. Then there exists a $g \in R$, $g \notin \mathfrak{p}$ such that the image of $x_1, \ldots, x_r$ in R_g forms an M_g-regular sequence.*

Proof. Set

$$K_i = \mathrm{Ker}\left(x_i : M/(x_1, \ldots, x_{i-1})M \to M/(x_1, \ldots, x_{i-1})M\right).$$

This is a finite R-module whose localization at $\mathfrak{p}$ is zero by assumption. Hence there exists a $g \in R$, $g \notin \mathfrak{p}$ such that $(K_i)_g = 0$ for all $i = 1, \ldots, r$. This g works. □

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